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August 27, 2026

To: Dr. Sahar Abdelnabi
Principal Investigator, COMPASS Research Group
ELLIS Institute Tübingen & Max Planck Institute for Intelligent Systems
Tübingen, Baden-Württemberg, Germany

Re: Application for Postdoctoral Researcher in AI Safety, Alignment, and Cooperative Agent Systems (COMPASS)

Dear Dr. Abdelnabi,

I am writing to submit my application for the Postdoctoral Researcher position within the COMPASS research group (COoperative Machine intelligence for People-Aligned Safe Systems) at the ELLIS Institute Tübingen and the Max Planck Institute for Intelligent Systems. As a final-year PhD candidate in Computer Science, my research investigates how to construct oversight signals that remain informative, ungameable, and robust when capable foundation models and multi-agent teams optimize against them.

De-Risking Oversight Signals: Anti-Reward-Hacking and Verifiable RL

In recent work (*Gah et al., under submission to SACAIR 2026*), I formulated an annotation-free process reward utilizing cross-view geometric consistency between frozen perception backbones. Rather than immediately training against the reward, I designed a two-arm validate-before-train protocol to test whether the supervision discriminated correct reasoning and whether it admitted exploitable shortcuts. This methodology surfaced a concrete specification gaming failure mode: predicted regions could exploit recall-heavy metrics by ballooning in spatial area without genuine precision. Redesigning the reward around count-based precision and weighted $F_{0.5}$ metrics reduced area bias correlation from $\rho > 0.72$ to 0.334, yielding a stable supervisory signal that optimized a Qwen2.5-VL-7B policy via online Group Relative Policy Optimization (GRPO) in VERL. At ELLIS, I aim to extend this counterfactual stress-testing methodology to multi-agent coding and reasoning systems.

Multi-Agent Anti-Collusion and Deception Detection

Consensus mechanisms face fundamental limitations in multi-agent environments because colluding agents can coordinate to satisfy surface-level oversight heuristics. My proposed research investigates mechanisms to detect deception and collusion using independent counterfactual checks, metamorphic mutation testing, and invariant sandboxing rather than superficial agent agreement. By analyzing agent communication through information-theoretic bottlenecks and causal probes, we can evaluate whether multi-agent coding pipelines remain aligned with intended specifications under joint reinforcement learning.

Calibrated Semantic Arbitration in Multi-Principal Environments

My doctoral thesis, *Reliability-Calibrated Adaptive Semantic Arbitration (RCASA)*, submitted to *IEEE TPAMI*, develops formal probabilistic calibration (Platt scaling, Bernoulli uncertainty modeling) to dynamically coordinate heterogeneous foundation models under distribution shift. This provides a mathematical foundation for multi-principal contextual integrity: arbitrating trust, resolving conflicting claims, and bounding information leakage when autonomous agents represent competing human principals with partially conflicting objectives.

Technical Implementation and Research Environment

My technical background includes hands-on experience in distributed RL frameworks (VERL, GRPO, vLLM), multi-agent orchestration (LangGraph, DeepEval, Opik), and fine-tuning pipelines. I look forward to contributing to COMPASS's research agenda and collaborating actively within the Tübingen AI and machine learning ecosystem.

Thank you for your time and consideration. I welcome the opportunity to discuss my research background and proposed work with you.

Sincerely,

Silas Kwabla Gah